Breast Cancer Prediction Using Machine Learning Models

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Problem Statement

- Breast cancer is a significant health challenge, being the most common cancer in women and a major cause of cancer-related deaths.
- The early detection of breast cancer plays a crucial role in treating infected patients.
  - Therefore, this project aims to utilize machine learning models to enhance breast cancer detection and diagnosis.

Project Motivation

- Improved Early Detection Machine learning enhances early detection of breast cancer.
- Accurate and Reliable Diagnosis - Advanced machine learning improves breast cancer diagnosis accuracy and reduces misdiagnosis rates.
- Integrating machine learning models into breast cancer prediction offers valuable research insights, aiding in a deeper understanding of the disease and advancing oncology.
- Decision Support for Healthcare Professionals: ML models can serve as decision support tools, assisting healthcare professionals in making more accurate and data-driven decisions in breast cancer diagnosis and treatment planning.

Project Delivery

- Utilize machine learning models to enhance breast cancer detection and diagnosis.
- The model can analyze diverse datasets, such as WBCD, for breast cancer info.
- The model can clearly distinguish between benign and malignant tumors, leading to more accurate breast cancer diagnoses.
- Continuous Learning: the model can continuously improve and adapt as more breast cancer data becomes available, ensuring up-to-date predictions.
- Design GUI Web Application to help detecting tumor cells if it is M or B.

Project Design and Structure

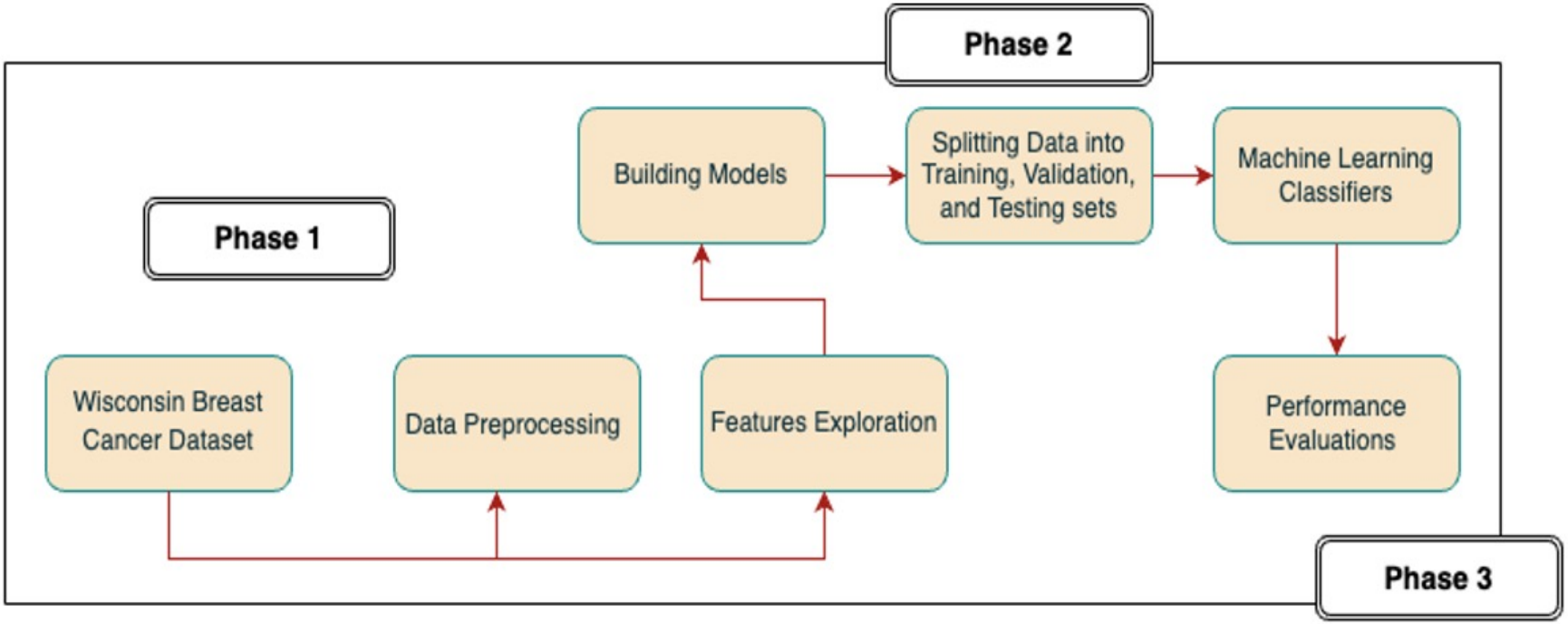


Figure 1: Workflow Diagram for Breast Cancer Detection Using Machine Learning Models

Project Design and data Info.

- The data acquired from Kaggle.
- The dataset has information calculated from a digitized images of breast cancer tumors, and it use to help for predicting if the tumor is benign or malignant.
- What are the features?

```
data.columns
```

```
Index(['diagnosis', 'radius_mean', 'texture_mean', 'perimeter_mean',
       'area_mean', 'smoothness_mean', 'compactness_mean', 'concavity_mean',
       'concave points_mean', 'symmetry_mean', 'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se', 'area_se', 'smoothness_se',
       'compactness_se', 'concavity_se', 'concave points_se', 'symmetry_se',
       'fractal_dimension_se', 'radius_worst', 'texture_worst',
       'perimeter_worst', 'area_worst', 'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave points_worst',
       'symmetry_worst', 'fractal_dimension_worst'],
      dtype='object')
```

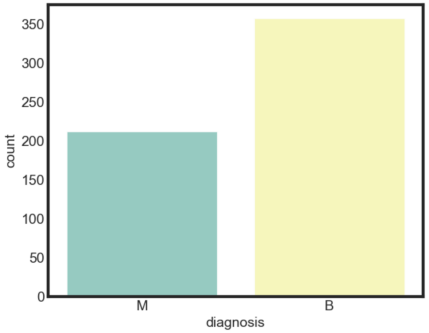
Project Design and data Info.

- Observation numbers and Productor Variable :

```
data.shape
```

```
(569, 31)
```

diagnosis	
B	357
M	212



- The dataset is spitted into 80 % for training and 20% for testing.

```
Y = data['diagnosis'].values
X = data.drop('diagnosis', axis=1).values
X_train, X_test, Y_train, Y_test = train_test_split (X, Y, test_size = 0.20, random_state=21)
```

```
X_train.shape , X_test.shape , Y_train.shape , Y_test.shape
```

```
((455, 30), (114, 30), (455,), (114,))
```

Project Design Decisions and Models

- Models were used:

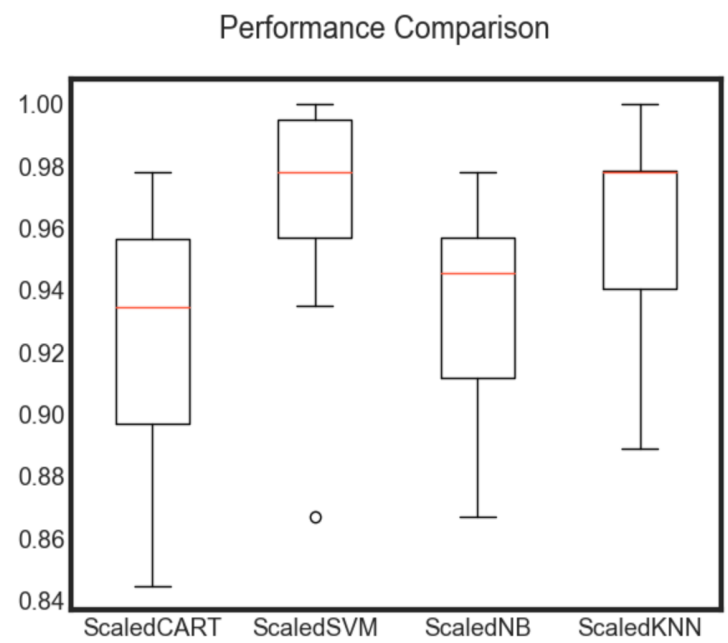
```
#Building Models
models_list = []
models_list.append(('CART', DecisionTreeClassifier()))
models_list.append(('SVM', SVC()))
models_list.append(('NB', GaussianNB()))
models_list.append(('KNN', KNeighborsClassifier()))
```

- Were there any particularly important design decisions ?

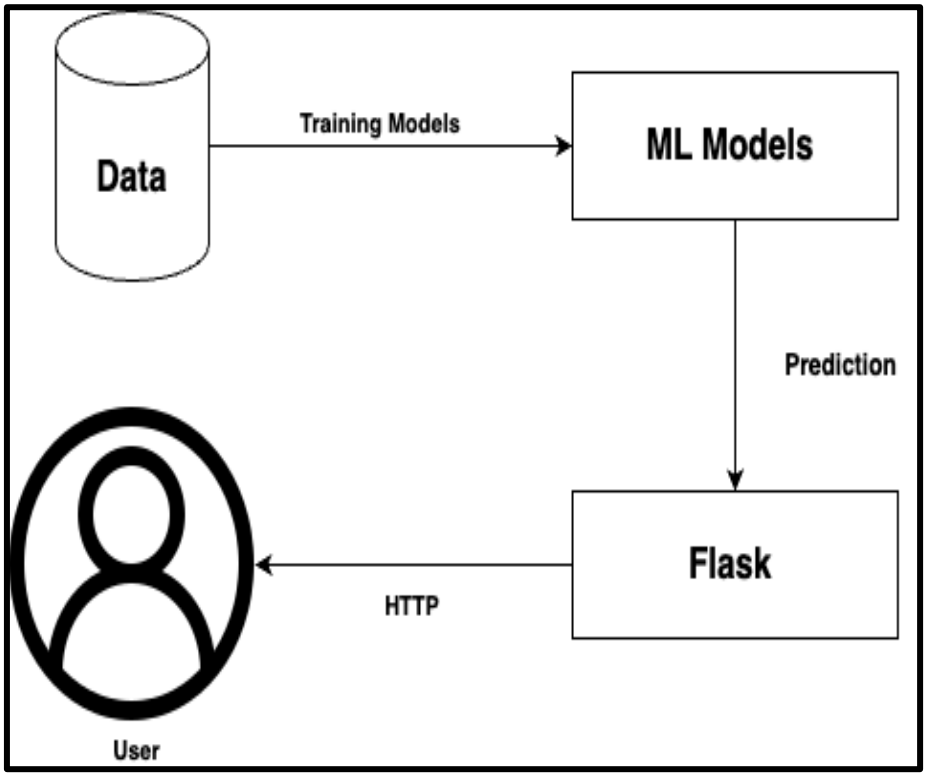
Based on the standardization process that has been made, SVM has the high accurate result. Therefore, I decided to optimize SVM to get the best results.

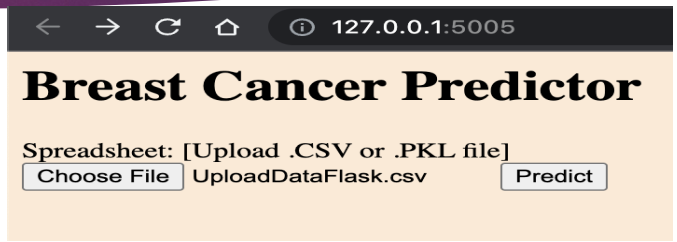
Comparison and Evaluation

Model	Precision	Recall	F-Score
CART	0.92	0.91	0.91
SVM	0.96	0.96	0.96
NB	0.93	0.93	0.92
KNN	0.96	0.95	0.95



Web Development and GUI





127.0.0.1:5005/predict

Data Information:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	1
0	10.860	21.48	68.51	360.5	0.07431	0.04227	0.00000	0.00000	0.1661	0.05948	0.3163	1.3040	2.1150	20.67	0.009579	0.011040	0.000000	0.000000	0.
1	14.950	18.77	97.84	689.5	0.08138	0.11670	0.09050	0.03562	0.1744	0.06493	0.4220	1.9090	3.2710	39.43	0.005790	0.048770	0.053030	0.015270	0.
2	20.290	14.34	135.10	1297.0	0.10030	0.13280	0.19800	0.10430	0.1809	0.05883	0.7572	0.7813	5.4380	94.44	0.011490	0.024610	0.056880	0.018850	0.
3	18.050	16.15	120.20	1006.0	0.10650	0.21460	0.16840	0.10800	0.2152	0.06673	0.9806	0.5505	6.3110	134.80	0.007940	0.058390	0.046580	0.020700	0.
4	23.510	24.27	155.10	1747.0	0.10690	0.12830	0.23080	0.14100	0.1797	0.05506	1.0090	0.9245	6.4620	164.10	0.006292	0.019710	0.035820	0.013010	0.
5	12.430	17.00	78.60	477.3	0.07557	0.03454	0.01342	0.01699	0.1472	0.05561	0.3778	2.2000	2.4870	31.16	0.007357	0.010790	0.009959	0.011200	0.
6	12.220	20.04	79.47	453.1	0.10960	0.11520	0.08175	0.02166	0.2124	0.06894	0.1811	0.7959	0.9857	12.58	0.006272	0.021980	0.039660	0.009894	0.
7	11.520	18.75	73.34	409.0	0.09524	0.05473	0.03036	0.02278	0.1920	0.05907	0.3249	0.9591	2.1830	23.47	0.008328	0.008722	0.013490	0.008670	0.
8	7.691	25.44	48.34	170.4	0.08668	0.11990	0.09252	0.01364	0.2037	0.07751	0.2196	1.4790	1.4450	11.73	0.015470	0.064570	0.092520	0.013640	0.
9	14.760	14.74	94.87	668.7	0.08875	0.07780	0.04608	0.03528	0.1521	0.05912	0.3428	0.3981	2.5370	29.06	0.004732	0.015060	0.018550	0.010670	0.

Predict Results

	Prediction
0	Benign
1	Benign
2	Malignant
3	Malignant

What’s next?

- Use different data set such as images.
- Test the data by using Deep Learning for enhancing the model's performance.

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DEMO.

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Thanks!

References

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